UNIVERZITET U BIHAĆU
TEHNIČKI FAKULTET
BIHAĆ

INTELIGENTNI SISTEMI

Laboratorijske vježbe

INVERZNE NEURONSKE MREŽE

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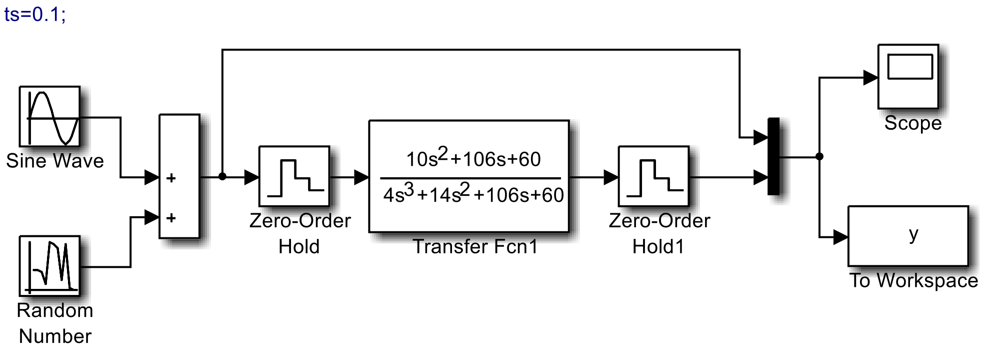
Inverzne neuronske mreže

PRIMJER 1: (*Obična neuronska mreža*) Kreirati neuronsku mrežu koja će oponašati funkciju:

$$\frac{10s^2 + 106s + 60}{4s^3 + 14s^2 + 106s + 60}$$



Slika 1: *Prvi model*

Podešavanja blokova:

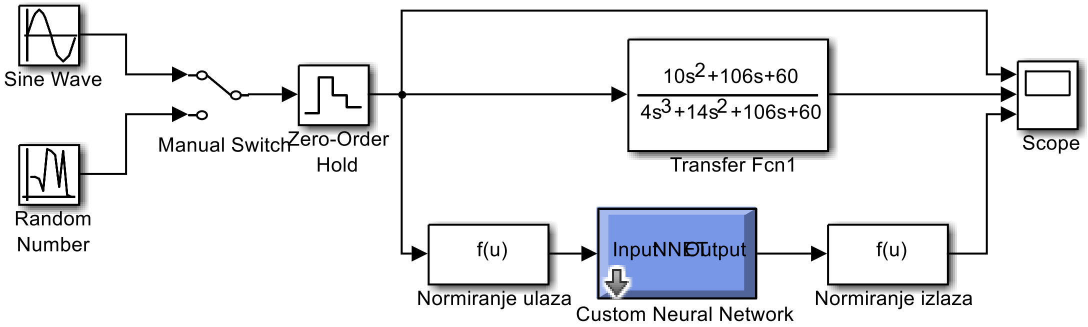
Sine Wave: *Amplitude (5), Bias (0), Frequency (0.1), Phase (0), Sample time (0)*

Random Number: *Mean (0), Variance (0.1), Seed (0), Sample time (ts)*

Zero-Order Hold: *Sample time (ts)*

Zero-Order Hold1: *Sample time (ts)*

Transfer Fcn1: *Num ([10 106 60]), Den ([4 14 106 60])*



Slika 2: *Drugi model*

Podešavanja blokova:

Sine Wave: *Amplitude (5), Bias (0), Frequency (0.1), Phase (0), Sample time (0)*

Random Number: *Mean (0), Variance (0.1), Seed (0), Sample time (ts)*

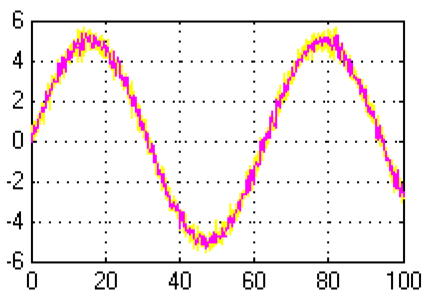
Zero-Order Hold: *Sample time (0.1)*

Zero-Order Hold1: *Sample time (ts)*

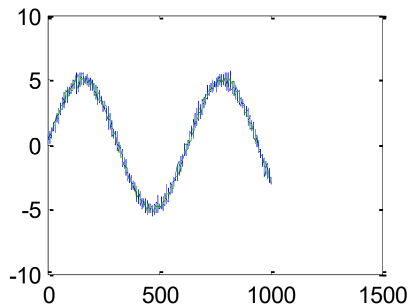
Normiranje ulaza: *Expression (2*(u(1)-minulaz)/(maxulaz-minulaz)-1), Sample time (0.1)*

Normiranje izlaza: *Expression ((u(1)+1)*(maxizlaz-minizlaz)/2+minizlaz), Sample time (0.1)*

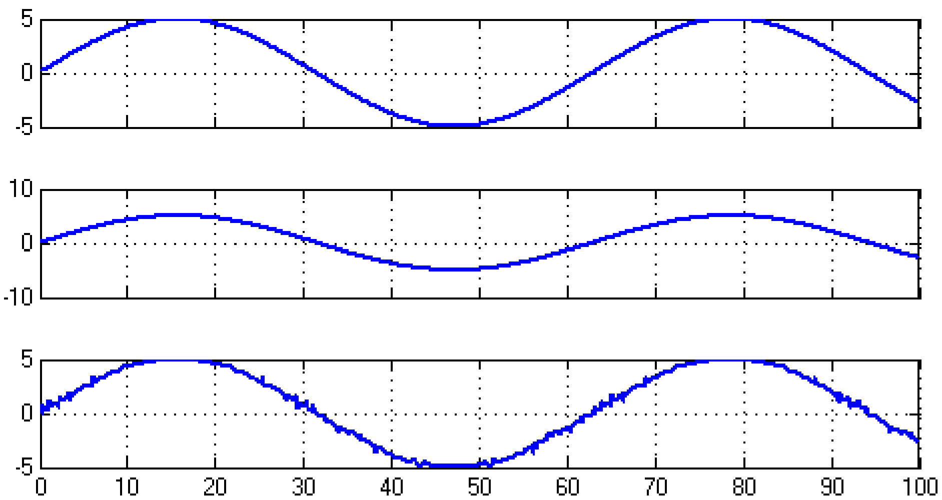
```
P=[y(:,1)];
T=[y(:,2)];
vel=length(P);
minulaz=min(P);
maxulaz=max(P);
minizlaz=min(T);
maxizlaz=max(T);
p=2*(P-minulaz)./(maxulaz - minulaz) - 1;      % normiranje ulaza
t=2*(T-minizlaz)./(maxizlaz - minizlaz) - 1;  % normiranje izlaza
net=newff([-1 1],[60 1],{'tansig','purelin'},'trainlm');
net.trainParam.epochs=1000;
net.trainParam.show=100;
net.trainParam.time=Inf;
net.trainParam.goal=2e-9;
net.performFcn='sse';
tic
net=train(net,p',t');
toc
izlaz=sim(net,p')
izlaz=(izlaz+1)*(maxizlaz-minizlaz)./2 +minizlaz;
figure
plot(P',izlaz,'r')
title('Podaci dobiveni sa treniranom mrezom');
xlabel('Uzorci')
ylabel('Amplituda')
gensim(net)
```



Slika 3: *Izlaz iz prvog sistema*



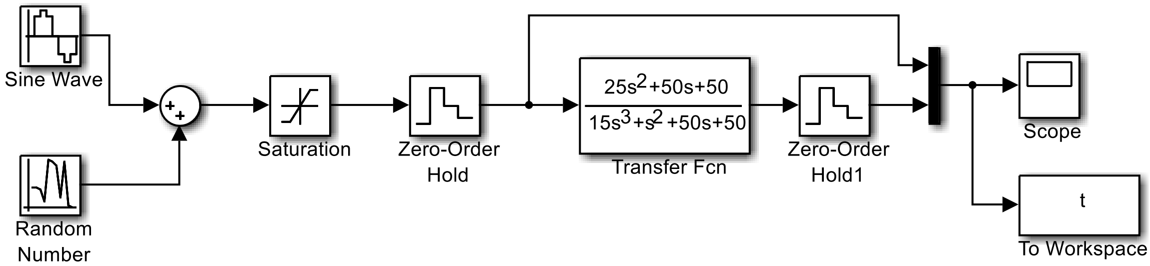
Slika 4: *Podaci dobiveni treniranjem NM*



Slika 5: *Podaci dobiveni na izlazu Simulink modela NM*

PRIMJER 2: (*Inverzna neuronska mreža*) Kreirati *neuronsku mrežu* koja će oponašati funkciju:

$$\frac{25s^2 + 50s + 50}{15s^3 + s^2 + 50s + 50}$$



Slika 6: *Prvi model*

Podešavanja blokova:

Sine Wave: *Amplitude (1), Bias (0), Frequency (1), Phase (0), Sample time (0.1)*

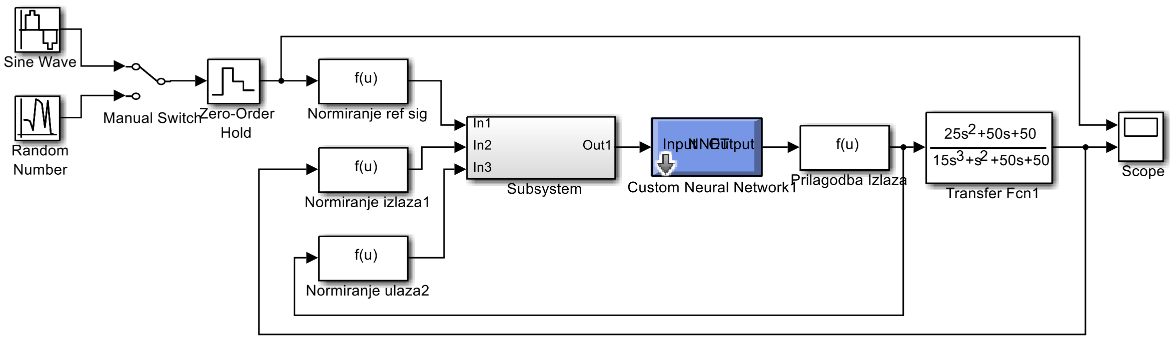
Random Number: *Mean (0), Variance (1), Seed (0), Sample time (0.1)*

Saturation: *Upper limit (0.5), Lower limit (-0.5), Sample time (0.1)*

Zero-Order Hold: *Sample time (0.1)*

Zero-Order Hold1: *Sample time (0.1)*

Transfer Fcn: *Num ([25 50 50]), Den ([15 1 50 50])*



Slika 7: *Drugi model*

Podešavanja blokova:

Sine Wave: *Amplitude (1), Bias (0), Frequency (1), Phase (0), Sample time (0.1)*

Random Number: *Mean (0), Variance (1), Seed (0), Sample time (0.1)*

Zero-Order Hold: *Sample time (0.1)*

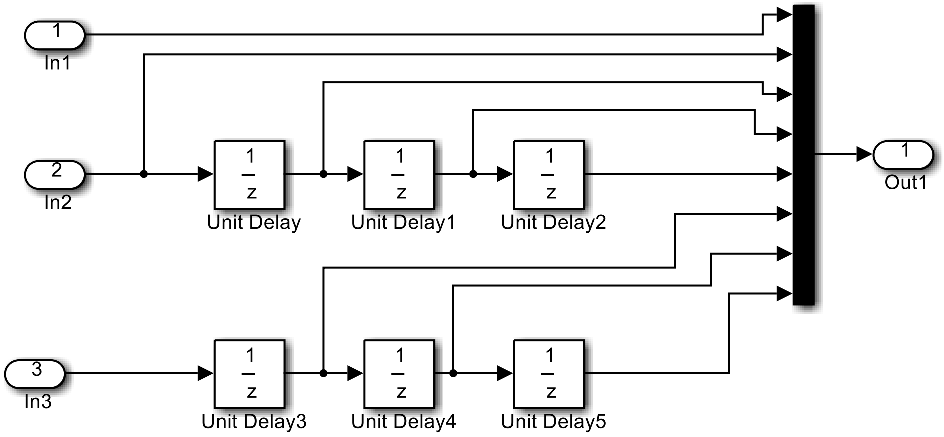
Zero-Order Hold1: *Sample time (ts)*

Normiranje ref.signala: *Expression (2*(u(1)-minizlaz)/(maxizlaz-minizlaz)-1), Sample time (0.1)*

Normiranje izlaza1: *Expression (2*(u(1)-minizlaz)/(maxizlaz-minizlaz)-1), Sample time (0.1)*

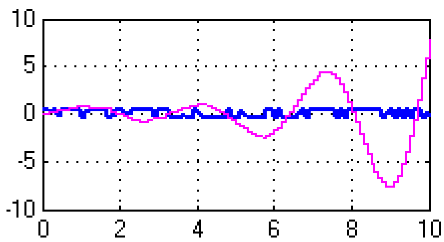
Normiranje ulaza2: *Expression (2*(u(1)-minulaz)/(maxulaz-minulaz)-1), Sample time (0.1)*

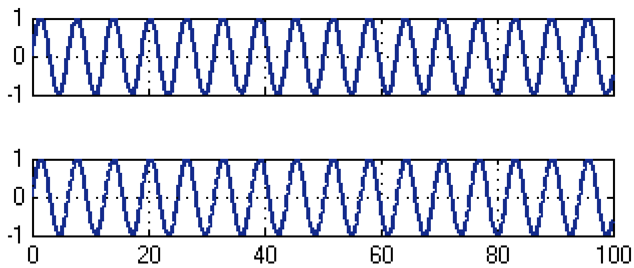
Prilagodba izlaza: *Expression ((u(1)+1)*(maxulaz-minulaz)/2+minulaz), Sample time (0.1)*



Slika 8: *Subsystem*

```
N=4;
P = t(:, 1);
T = t(:, 2);
minulaz = min(T); maxulaz = max(T);
minizlaz = min(P); maxizlaz = max(P);
net = newff([zeros(2*N,1)-1 zeros(2*N,1)+1], ...
[15 5 1],{'tansig','tansig', 'purelin'}, 'trainlm');
net.trainParam.epochs = 2000;
net.trainParam.goal=2e-9;
net.trainParam.show = 10;
net.trainParam.time = Inf;
net.performFcn = 'sse';
P= 2 * (P - minulaz) ./ (maxulaz - minulaz) - 1;
T = 2 * (T - minizlaz) ./ (maxizlaz - minizlaz) - 1;
vel = length(T);
ulaz = zeros(2*N, vel-N);
izlaz = zeros(1, vel-N);
for k = N : vel-1
t = flipud( T(k-N+1:k+1) );
p = flipud( P(k-N+1:k-1) );
ulaz(:,k) = [t; p];
izlaz(k) = P(k);
end;
 tic
net = train(net, ulaz, izlaz);
 toc
izlaz=sim(net,ulaz);
izlaz=(izlaz+1)*(maxulaz-minulaz)./2 +minulaz;
gensim(net);
```

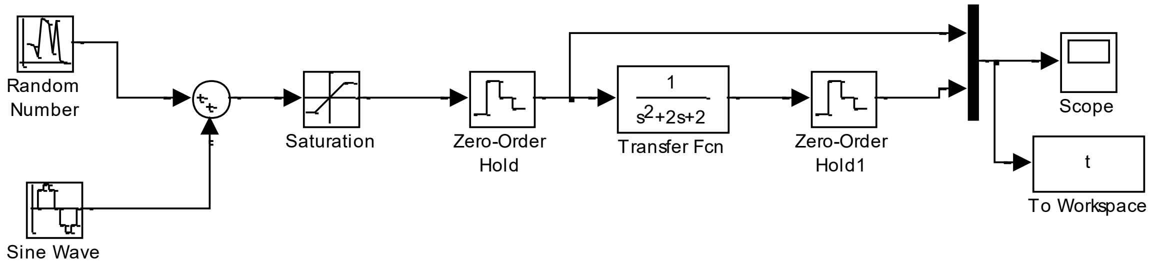




Slika 9: *Izlaz iz prvog sistema* **Slika 10:** *Podaci dobiveni na izlazu Simulink modela NM*

PRIMJER 3: (*Inverzna neuronska mreža*) Kreirati *neuronsku mrežu* koja će oponašati funkciju:

$$\frac{1}{s^2 + 2s + 2}$$



Slika 11: *Prvi model*

Podešavanja blokova:

Sine Wave: *Amplitude (4), Bias (0), Frequency (0.025), Phase (0), Sample time (0.1)*

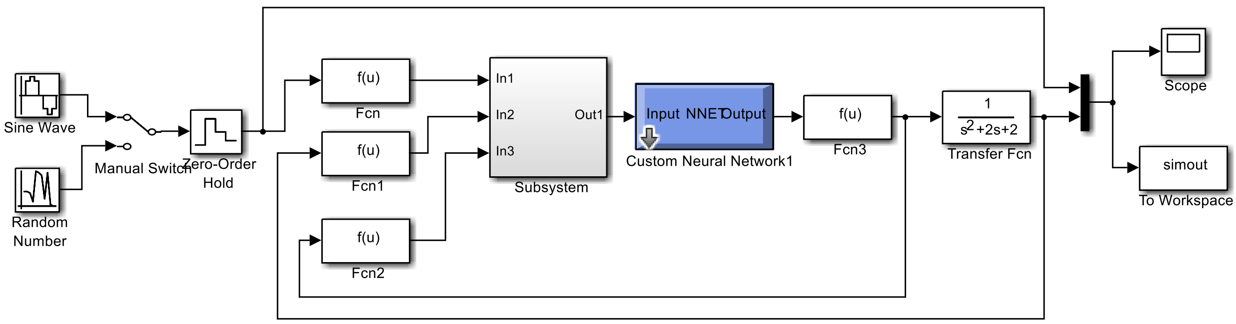
Random Number: *Mean (0), Variance (0.1), Seed (0), Sample time (0.1)*

Saturation: *Upper limit (5), Lower limit (-5), Sample time (0.1)*

Zero-Order Hold: *Sample time (0.1)*

Zero-Order Hold1: *Sample time (0.1)*

Transfer Fcn: *Num ([1]), Den ([1 2 2])*



Slika 12: *Drugi model*

Podešavanja blokova:

Sine Wave: *Amplitude (4), Bias (0), Frequency (0.025), Phase (0), Sample time (0.1)*

Random Number: *Mean (0), Variance (0.1), Seed (0), Sample time (0.1)*

Zero-Order Hold: *Sample time (0.1)*

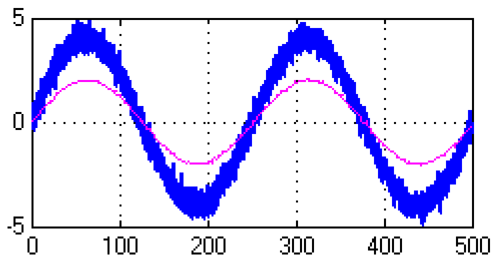
Normiranje ref.signala (Fcn): *Expression (2*(u(1)-minizlaz)/(maxizlaz-minizlaz)-1), Sample time (0.1)*

Normiranje izlaza1 (Fcn1): *Expression (2*(u(1)-minizlaz)/(maxizlaz-minizlaz)-1), Sample time (0.1)*

Normiranje ulaza2 (Fcn2): *Expression (2*(u(1)-minulaz)/(maxulaz-minulaz)-1), Sample time (0.1)*

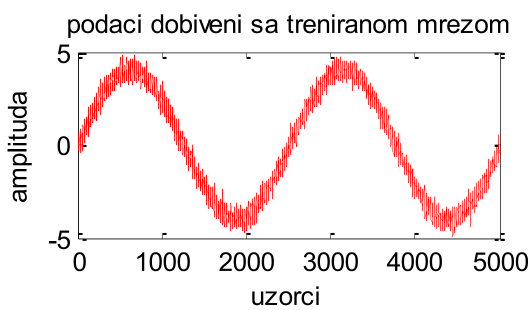
Prilagodba izlaza (Fcn3): *Expression (((u(1)+1)*(maxulaz-minulaz)/2+minulaz), Sample time (0.1)*

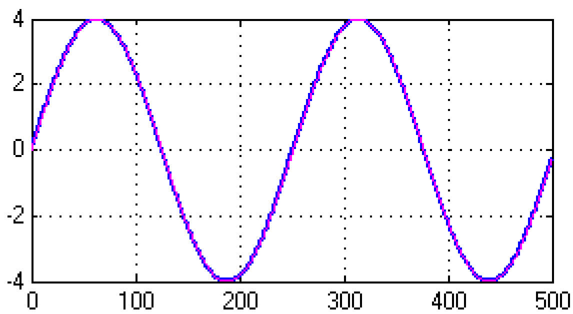
```
N=4;   %4  ulaza u mrezu
broj_epoha=500;
minulaz=-3;
maxulaz=3; %opseg ulaznih vrijednosti
minizlaz=-2.2;
maxizlaz=2.2;
net=newff([zeros(2*N,1)-1 zeros(2*N,1)+1],... %svi moraju biti na intervalu od -1 1
    [15 5 1],{'tansig','tansig','purelin'},'trainlm');
fprintf('Pocetak treniranja \n');
net.trainParam.epochs=broj_epoha; %trenira mrežu 500 puta
net.trainParam.goal=0;
net.trainParam.show=10;  % svaki deset puta prikazuje rezultat
net.trainParam.time=Inf;
fprintf('opseg ulaza mreze je :[%g,%g].\n',minizlaz,maxizlaz);
fprintf('priprema podataka za treniranje mreze...\n');
P=t(:,1);
T=t(:,2);
%normiranje ulaza i izlaza na opseg [-1 1]
P=2*(P-minulaz)./(maxulaz-minulaz)-1;
T=2*(T-minizlaz)./(maxizlaz-minizlaz)-1;
vel=length(T);
ulaz=zeros(2*N,vel-N);
izlaz=zeros(1,vel-N);
for k=N:vel-1
    t=flipud(T(k-N+1:k+1)); % okrece niz,zadnji dio stavlja na prvi
    p=flipud(P(k-N+1:k-1));
    ulaz(:,k)=[t;p];
    izlaz(k)=P(k);
end
fprintf('pocetak treniranja\n');
tic
net=train(net,ulaz,izlaz);
toc
```



Slika 13: *Izlaz iz prvog sistema*

```
izlaz=sim(net,ulaz);
izlaz=(izlaz+1)*(maxulaz-minulaz)./2+minulaz; %vraca iz normiranog u stvarni oblik
figure
plot(izlaz,'r');
title('podaci dobiveni sa treniranom mrezom')
xlabel('uzorci');
ylabel('amplituda')
gensim(net,0.1)
```





Slika 14: *Podaci dobiveni treniranjem NM* **Slika 15:** *Podaci dobiveni na izlazu Simulink modela NM*